

For Problem Statements 1-4:

Submission Instructions:

- Submit a working backend demo (local with ngrok or simple cloud instance).
- Provide a GitHub repository containing your code.
- Include a README.md with setup instructions, architecture, and API usage examples.
- Provide sample curl/Postman requests to test your endpoints.
- Submit a short screen recording showing your endpoints working with sample inputs.

Evaluation Criteria:

- Correctness of API responses and adherence to JSON schemas.
- Handling of both text and image inputs with OCR.
- Implementation of guardrails and error handling.
- Code organization, clarity, and reusability.
- Effective use of AI for chaining and validation.

Problem Statement 3: AI-Powered Medical Report Simplifier

Focus Area: OCR -> Test Extraction -> Plain-Language Explanation

Problem Statement:

Create a backend service that takes medical reports (typed or scanned) and produces patient-friendly explanations. Interns must handle OCR errors, normalize tests, and ensure no hallucinated results are added. Final output must include normalized tests and simplified explanations.

Step 1 - OCR/Text Extraction

Extract test names, values, units; fix minor typos.

Input (text):

CBC: Hemoglobin 10.2 g/dL (Low), WBC 11,200 /uL (High)

Input (image -> OCR sample):

CBC: Hemoglobin 10.2 g/dL (Low)
WBC 11200 /uL (Hgh)

Expected Output (JSON):

```
{
  "tests_raw": [
    "Hemoglobin 10.2 g/dL (Low)",
    "WBC 11200 /uL (High)"
  ],
  "confidence": 0.80
}
```

Step 2 - Normalized Tests JSON

Standardize names, units, ranges, and statuses.

Expected Output (JSON):

```
{
  "tests": [

    {"name": "Hemoglobin", "value": 10.2, "unit": "g/dL", "status": "low",
    "ref_range": {"low": 12.0, "high": 15.0}},

    {"name": "WBC", "value": 11200, "unit": "/uL", "status": "high", "ref_
    range": {"low": 4000, "high": 11000}}

  ],
  "normalization_confidence": 0.84
}
```

Step 3 - Patient-Friendly Summary

Explain findings simply without diagnosing.

Expected Output (JSON):

```
{
  "summary": "Low hemoglobin and high white blood cell count.",
  "explanations": [
    "Low hemoglobin may relate to anemia.",
    "High WBC can occur with infections."
  ]
}
```

Guardrail / Exit Condition (JSON):

```
{"status": "unprocessed", "reason": "hallucinated tests not present
in input"}
```

Step 4 - Final Output

Return combined normalized tests and summary.

Expected Output (JSON):

```
{
  "tests": [

    {"name": "Hemoglobin", "value": 10.2, "unit": "g/dL", "status": "low",
    "ref_range": {"low": 12.0, "high": 15.0}},

    {"name": "WBC", "value": 11200, "unit": "/uL", "status": "high", "ref_
    range": {"low": 4000, "high": 11000}}

  ],
  "summary": "Low hemoglobin and high white blood cell count.",
  "explanations": [
    "Low hemoglobin may relate to anemia.",
    "High WBC can occur with infections."
  ]
}
```

```
],  
  "summary": "Low hemoglobin and high white blood cell count.",  
  "status": "ok"  
}
```